Predicting Air Pressure System (APS) Failures in Scania Trucks

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1. Abstract and Introduction

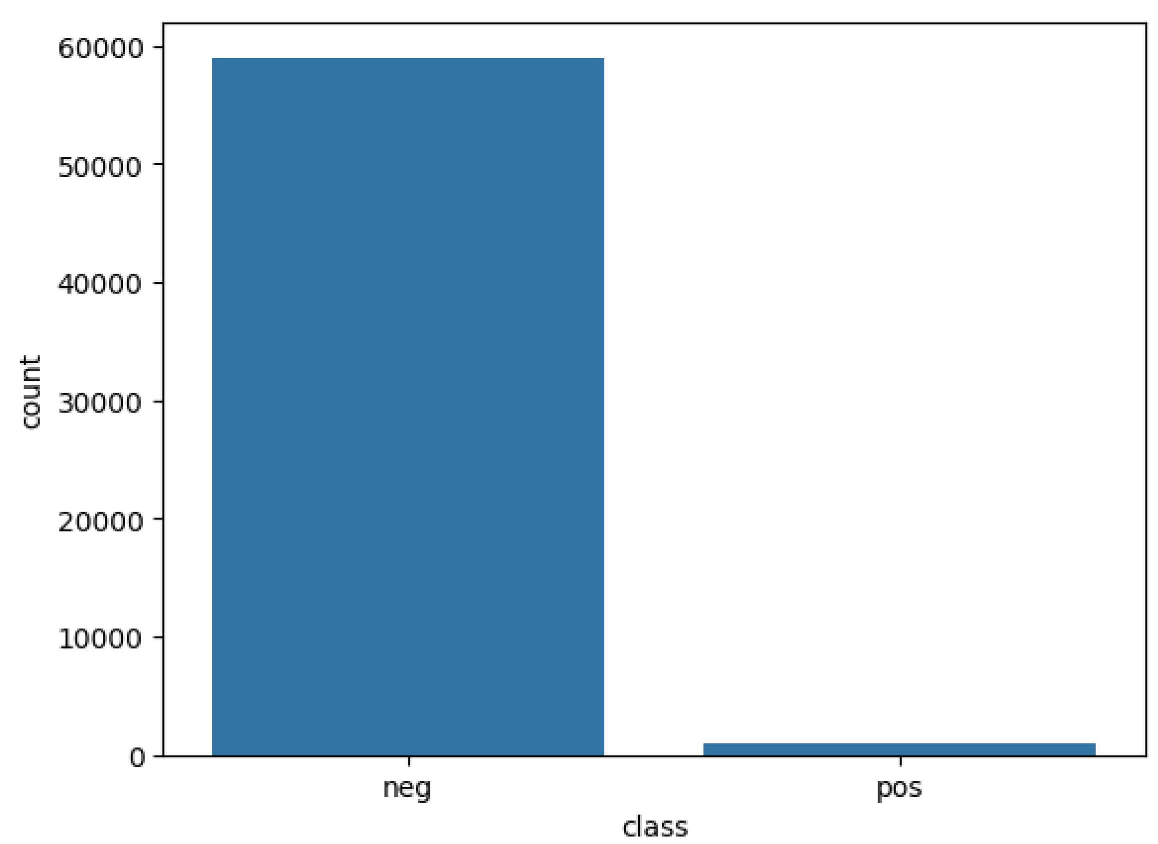
The Air Pressure System (APS) is essential for the braking and gear shifting system in Scania trucks. Undetected failures can lead to costly repairs, vehicle downtime, and safety risks. This project uses sensor data and machine learning models to predict APS failures before they occur.

Objectives

- Perform Exploratory Data Analysis (EDA) to understand the dataset
- Preprocess and prepare the data for modeling
- Train and evaluate six machine learning models
- Identify the model with the best balance of precision and recall (F1-score)
- Provide insights for predictive maintenance applications

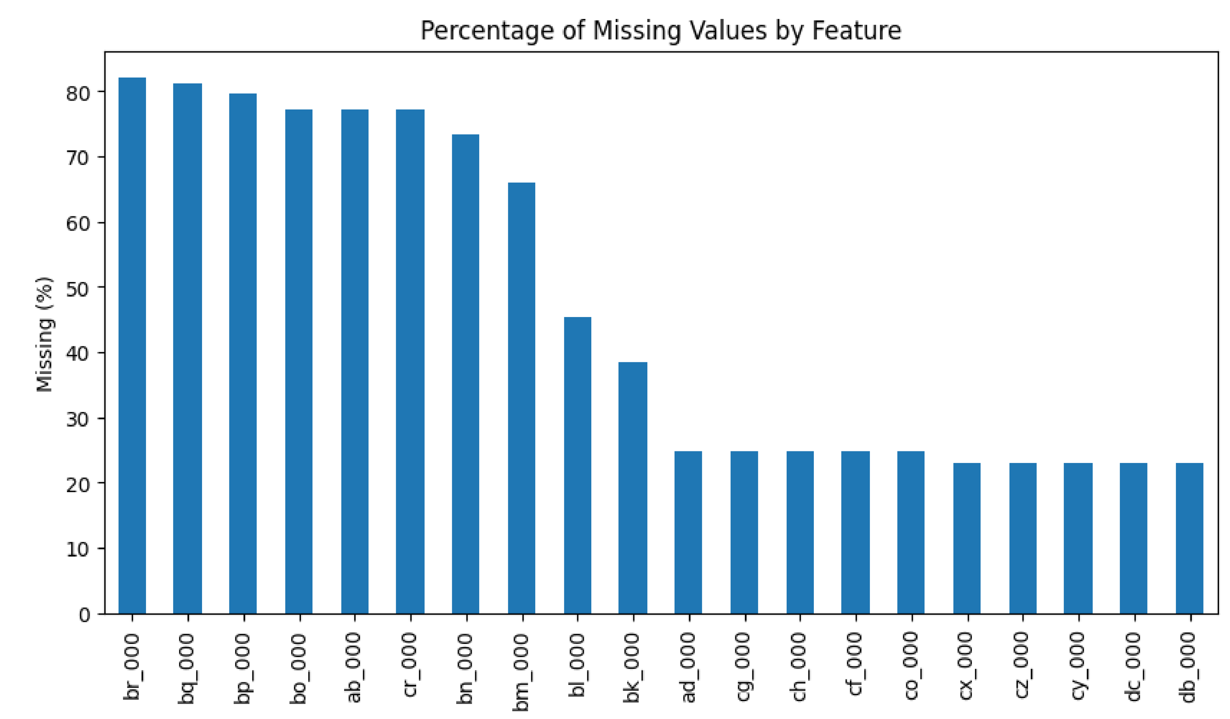
2. Exploratory Data Analysis (EDA)

2.1 Class Distribution



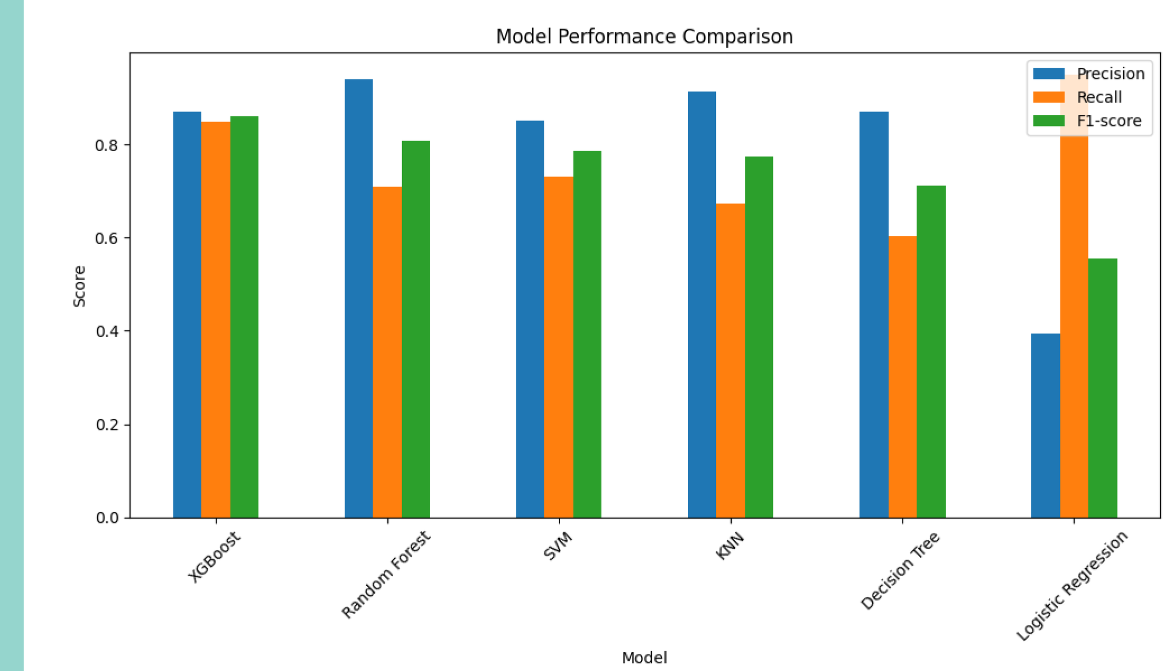
Extreme class imbalance: only 1.67% of instances are APS failures

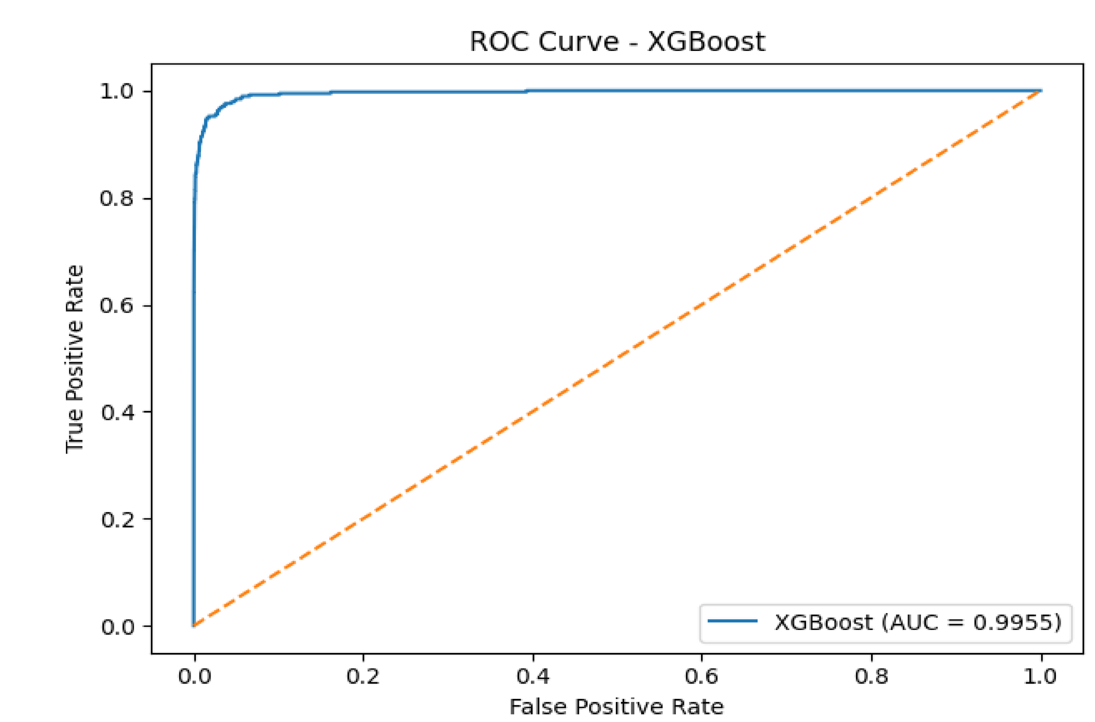
2.2 Missing Values



Several features had >80% missing values and were removed

5. Results





3. Methodology

3.1 Preprocessing Pipeline

- Replace "na" with NaN
- Convert data types to numeric
- Remove features with >80% missing values
- Median imputation
- Log1p transformation
- Standardization (StandardScaler)

3.2 Machine Learning Models (6)

- Logistic Regression
- Decision Tree
- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)
- Random Forest
- XGBoost

3.3 Hyperparameter Tuning

GridSearchCV with cross-validation (F1-score as optimization metric)

3.4 Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-score (primary metric)

Methodology (cont.) – Experimental Setup

- Train/Test split provided
- All preprocessing fitted on training set and applied to test set (no data leakage)
- Implemented in Python using:
  - Pandas, NumPy, scikit-learn, XGBoost, Matplotlib, Seaborn

Model Performance Summary (Test and Training Datasets)

Rank (by F1 score)	Model	Train Accuracy	Train Precision	Train Recall	Train F1-score	Test Accuracy	Test Precision	Test Recall	Test F1-score
5	XGBoost	0.993500	0.871233	0.848000	0.859459	0.993500	0.871233	0.848000	0.859459
4	Random Forest	0.992125	0.939929	0.709333	0.808511	0.992125	0.939929	0.709333	0.808511
3	SVM	0.990688	0.850932	0.730667	0.786227	0.990688	0.850932	0.730667	0.786227
2	KNN	0.990812	0.913043	0.672000	0.774194	0.990812	0.913043	0.672000	0.774194
1	Decision Tree	0.988563	0.869231	0.602667	0.711811	0.988563	0.869231	0.602667	0.711811
0	Logistic Regression	0.964500	0.393370	0.949333	0.556250	0.964500	0.393370	0.949333	0.556250

XGBoost achieved the highest F1-score (0.8595), demonstrating the best balance between precision and recall on the Imbalanced APS dataset.

Conclusion

Key Takeaways

- Severe class imbalance (1.67%) requires careful handling appropriate metrics
- Ensemble methods outperform traditional methods for this classification task.
- XGBoost provides the best balance of precision and recall, minimizing missing failures while keeping false positive low.

Future Work

- Expand hyperparameter search for better optimization.
- Evaluate models on additional or more recent datasets to test generalization.
- Explore additional feature engineering technique and alternative models.

Answers to Research Questions

Q1: Which model provides the best overall performance?

- XGBoost had the highest F1-score (0.8595) and best balance of precision and recall.

Q2: How does class imbalance affect performance?

- Without adjusting class weights, models are biased towards the majority class, causing low recall. Adjusting the class weights improved failure detection.

Q3: What preprocessing steps and modeling techniques are most important?

- Median imputation, log transformation, and standardization were most important. Random Forest and XGBoost handled the dataset imbalance and complexity the best.